

MLCC 2026

Lab 4/5 – Nonlinear models

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Nonlinear models

Hypothesis space $\mathcal{H}$ = representation/parametrization of the functions

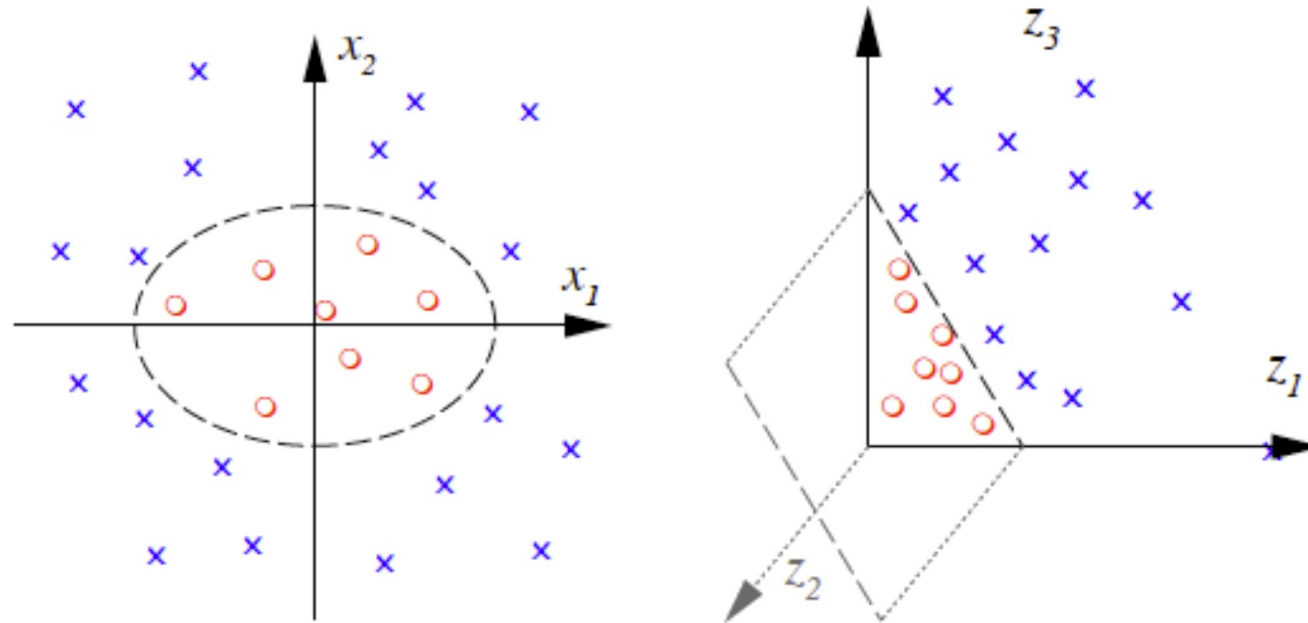
$$\underbrace{w^T x}_{\text{linear}} \qquad \underbrace{w^T \Phi(x) \quad w^T \sigma(W^T x)}_{\text{nonlinear}}$$

Nonlinear models

$$w^T \Phi(x)$$

$$\Phi : R^2 \rightarrow R^3$$

$$(x_1, x_2) \mapsto (z_1, z_2, z_3) := (x_1^2, \sqrt{2}x_1x_2, x_2^2)$$



Nonlinear models

Kernels

$$f(x) = w^T \Phi(x) \rightarrow f(x) = \sum_{i=1}^n c_i k(x_i, x) \quad \left(k(x, x') = \Phi^T(x) \Phi(x') \right)$$

Examples:

- Linear
- Polynomial
- Gaussian

$$k(x, x') = x^T x'$$

$$k(x, x') = (x^T x' + 1)^d$$

$$k(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\sigma^2}\right)$$

Nonlinear models

How to find c_i :

$$\hat{K}_{ij} = k(x_i, x_j), \quad i, j = 1, \dots, n$$

kernel_matrix

$$(\hat{K} + n\lambda I)c = \hat{Y}$$

$$\hat{c} = (\hat{K} + n\lambda I)^{-1} \hat{Y}$$

krls_train

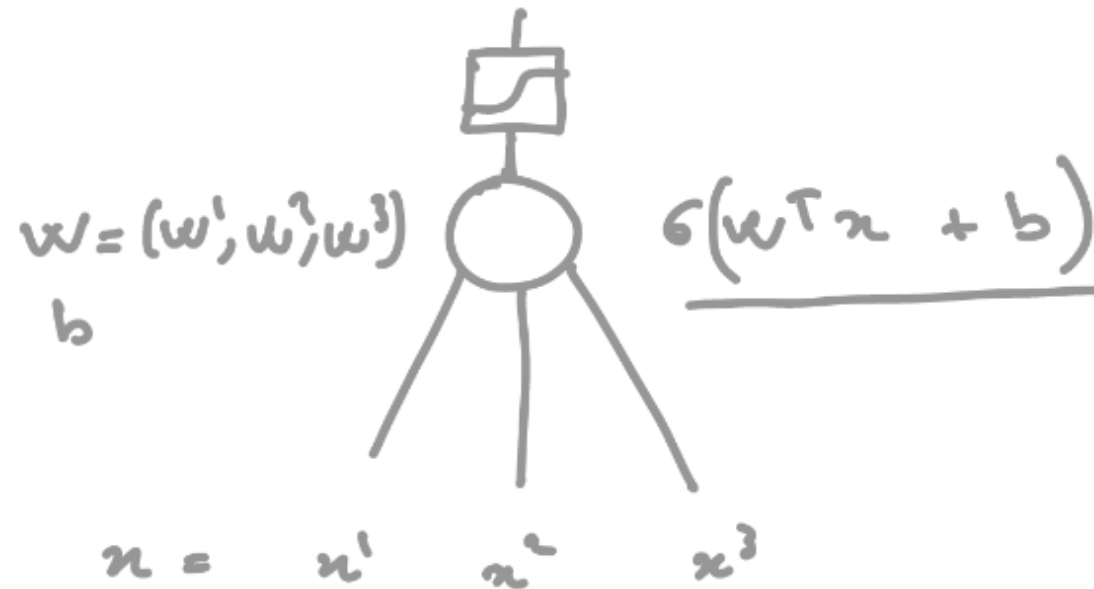
$$f(x) = \sum_{i=1}^n c_i k(x_i, x)$$

krls_predict

Kernels

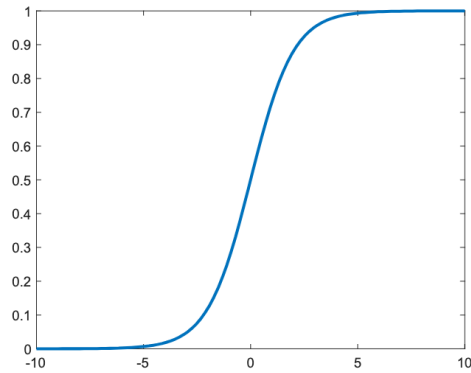
- Generate data for a nonlinear problem
- Try a linear model
- Try a feature map adapted to the data
- Try a Gaussian kernel

Neural networks

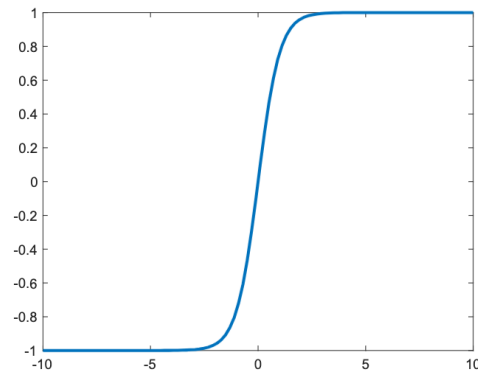


Neural networks

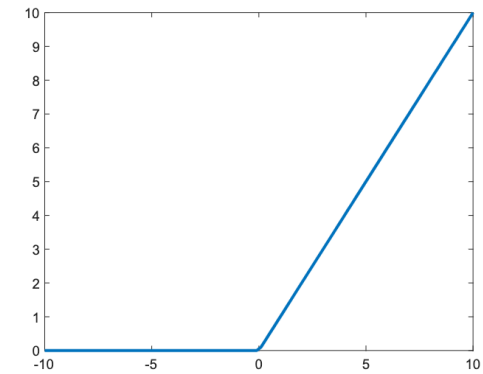
Activation functions



$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



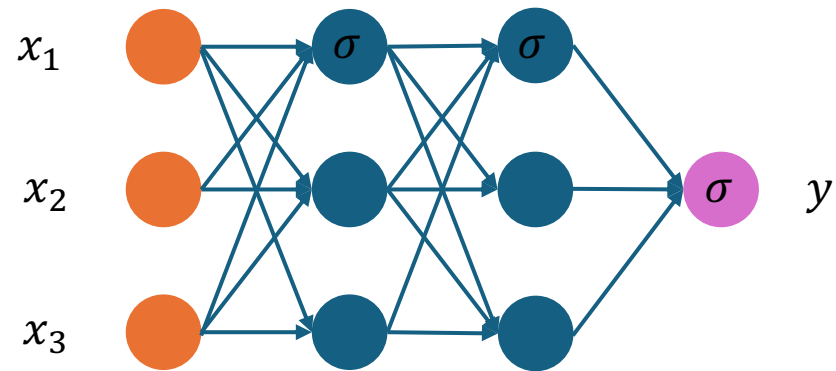
$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



$$f(x) = \max(0, x)$$

Neural networks

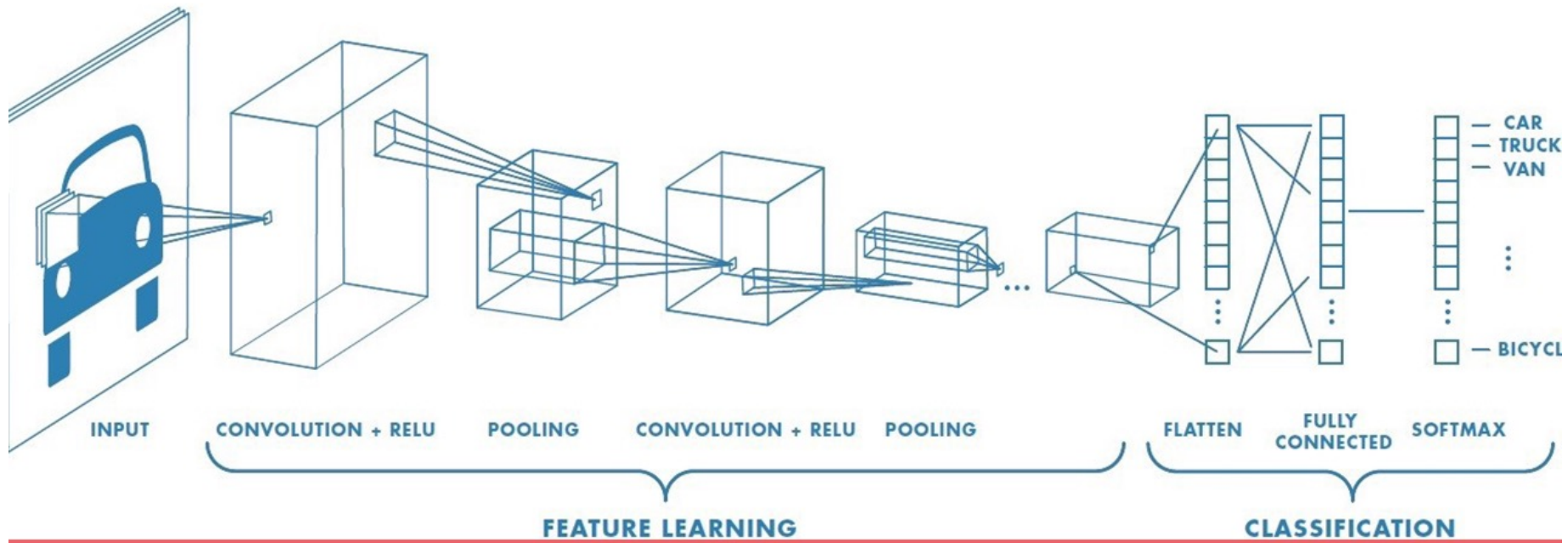
Multiple layers



Neural networks

Convolutional Neural Networks

Convolutional Neural Networks CNN PowerPoint Template



Neural Networks

- Nonlinear synthetic data
- Using Keras, explore different architectures and check how the decision boundary changes
- Binary classification with MNIST

BONUS: convolutions